Assignment-1 AIML Lab (1st lab practice qn)

1. Write a python program to Swap Two Numbers

```
a = 10
b = 20

a, b = b, a

print("a =", a)
print("b =", b)
```

Output:

```
a = 20
b = 10
```

2. Program to check a given Number is Prime or not

```
n = 7

if n > 1:
   for i in range(2, n):
      if n % i == 0:
         print("Not Prime")
         break
   else:
      print("Prime")
else:
   print("Not Prime")
```

Output:

```
Prime
```

3. Python program to find Nth Fibonacci Number

```
n = 7
a = 0
b = 1

for i in range(n):
   a, b = b, a + b

print("Fibonacci number =", a)
```

Output:

```
Fibonacci number = 13
```

4. Multiplication Table 1-n

```
n = 5

for i in range(1, 11):
   print(n, "x", i, "=", n * i)
```

Output:

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

5. Inverted Pyramid Pattern

```
n = 5

for i in range(n, 0, -1):
   print("* " * i)
```

Output:

```
* * * * *
* * * *
* * *
* *
*
```

6. Right-Angled Triangle Pattern

```
n = 5

for i in range(1, n + 1):
   print("* " * i)
```

Output:

```
*
* *
* * *
* * * *
* * * * *
```

7. Greatest of 3 Numbers

```
a = 10
b = 25
c = 15

if a > b and a > c:
   print("Greatest =", a)
elif b > c:
   print("Greatest =", b)
else:
   print("Greatest =", c)
```

Output:

Greatest = 25

8. Display even Numbers Between a Range

```
start = 1
end = 10

for i in range(start, end + 1):
   if i % 2 == 0:
      print(i)
```

Output:

2
4
6
8
10

9. Factorial of a Number

```
n = 5
fact = 1

for i in range(1, n + 1):
   fact = fact * i

print("Factorial =", fact)
```

Output:

Factorial = 120

10. Check Odd or Even Using Simple if

```
n = 8

if n % 2 == 0:
   print("Even")
else:
   print("Odd")
```

Output:

```
Even
```